A LOW-CODE, NO-CODE
INFORMATICS PLATFORM

Deploying Configurations between Sapio Environments Guide

The Sapio platform offers powerful Out-of-the-Box (OOTB) features, which can be customized by customers, their Sapio team, or partners. These custom configurations can be seamlessly exported and imported across Sapio environments using Synergies.

Table of Contents

Deploying Configurations between Sapio Environments Guide 1
Table of Contents 1
What is a Synergy? 2
- What can be Exported/Imported with the Export/Import Synergy Tools? 3
- What CANNOT be Exported/Imported with the Export/Import Synergy Tools? 3
- What Can Be Exported/Imported with the Sapio Configurations Tool? 3
- What CANNOT Be Exported/Imported with the Sapio Configurations Tool? 4

Flow of Deployment between Sapio Environments 5
- Export General System Settings with a Synergy 6
- Export Sapio Configurations 7
- Import General System Settings with a Synergy 9
- Import Sapio Configurations 11
- Update Webhook Base URLs following synergy import of Webhooks 13

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620

This document is **RESTRICTED**

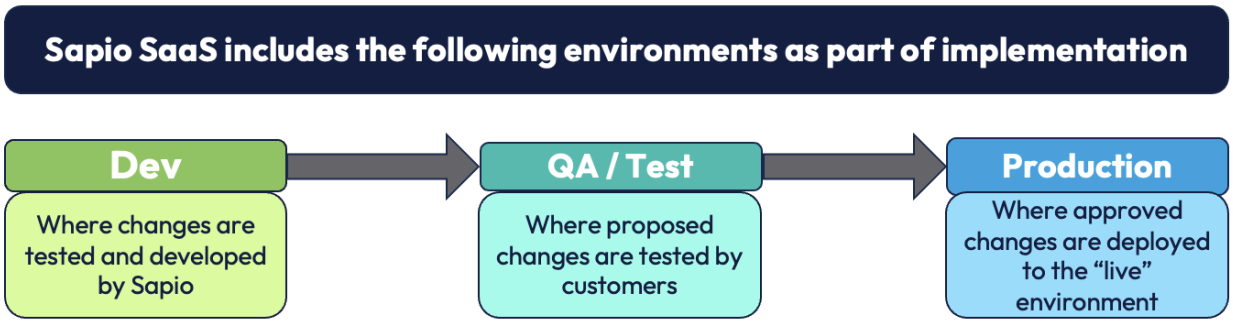
What is a Synergy?

A Synergy is a deployment utility file (.ssg) generated by Sapio, enabling administrators to quickly transfer design and data structure settings between environments with a simple download/upload. Synergies streamline iterative design updates throughout the development cycle.

The Sapio development cycle flows through three environments, or instances, that each customer gets as a part of their product.

The three environments represent distinct URLs and serve different purposes...

- **Dev Environment :** Where the Implementation Team builds and tests custom solutions.
- **Test or QA Environment :** Where customers conduct User Acceptance Testing (UAT) before deployment.
- **Production :**The live environment where real data is processed and stored.



Note : On-Premise or legacy accounts may have different environment structures

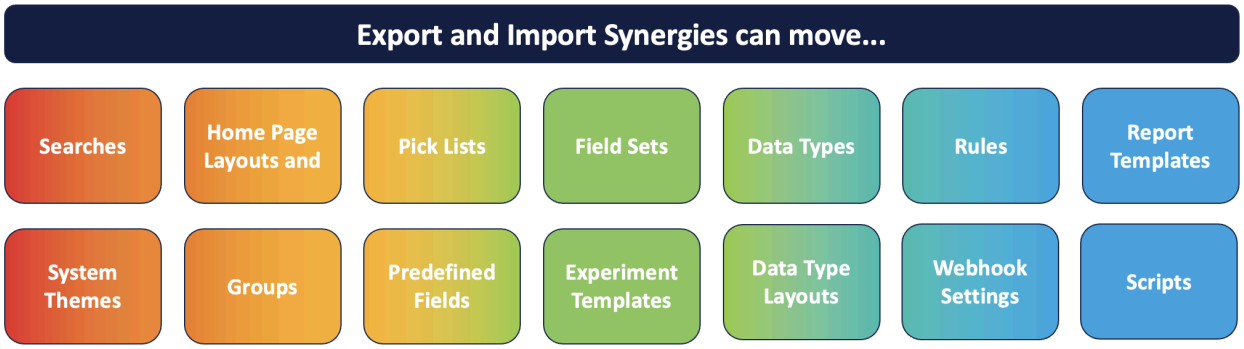
Once the development for a given cycle finishes, the configurations should be moved using synergies. Configurations are moved at two levels by export/import in Sapio...

- **General System Configurations :** exported/imported using the Export/Import Synergy tools in the App Setup toolbar
- **Sapio Configurations :** exported/imported using the 'Import/Export Sapio Configurations' tool in Configuration Manager of App Setup

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

What can be Exported/Imported with the Export/Import Synergy Tools?

The following general system configurations can be moved by Export/Import Synergy : Searches, Home Page Layouts, Pick Lists, Field Sets & Predefined Fields, Data Types, Rules, Report Builder Templates, Chart Definitions, System Themes, Groups, Predefined Fields, Experiment Templates, Protocol Templates, Data Type Layouts, Webhook Settings, and Scripts, and Corrections.



What CANNOT be Exported/Imported with the Export/Import Synergy Tools?

Synergies will not export or import...

- Record Data
- User assignments to groups

Synergies will also not delete fields or remove relationships (this must be done manually).

Tip : Contact Sapio Support to transfer record data

What can be Exported/Imported with the Sapio Configurations Tool?

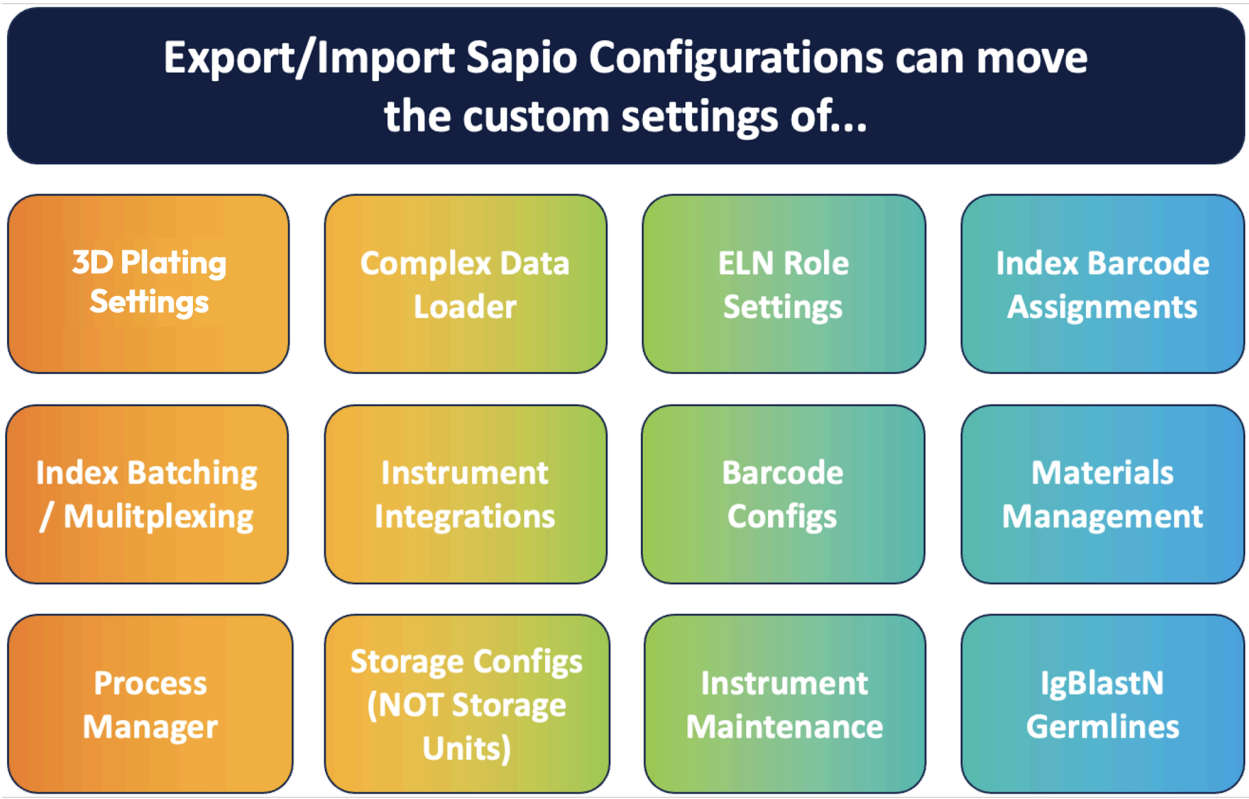
Sapio Configurations are a collection of data records that store customizations specific to your Sapio application. These are managed using "Configuration Manager" of App Setup. Sapio Configurations differ from synergies by exporting only settings defined under Configuration Manager instead of general system settings

The following components are examples of what is exported when using the 'Import/Export Sapio Configurations' tool under Configuration Manager in App Setup: 3D

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

Plating Templates, Complex Data Loader, ELN Role Settings, Index Barcode Assignments, Index Batching/Multiplexing, Instrument Integrations, Barcode Configs, Materials Management settings, Processes (including column definitions for the queues), Storage Configurations, Instrument Maintenance settings, and igBlastN Germlines.

Note : Individual Storage units must still be created and are not transferred by synergy



Tip: 3D plating templates are transferred by synergy as a part of Experiment Templates. Legacy plate mapping Templates ARE transferred by Sapio Configurations.

What CANNOT Be Exported/Imported with the Sapio Configurations Tool?

Sapio Configurations will not export or import...

- Storage Units
- Reagents
- Instruments

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

Tip: Contact Sapio support if you have a large volume of non transferable data to review alternative transfer options.

Flow of Deployment between Sapio Environments

Configurations should be deployed in a particular order to ensure seamless export and import.

The export should include:

1. Creating a branch for all existing code in the environment (for example, git)

2. Have an admin go to App Setup > Export Synergy and select all the system wide settings/data to export (for example, added Data Types, Experiment Templates, etc.)

Tip: Ideally, you should be able to select "Export All Configurations" and select "Latest Versions" to get a full copy. The only reason to "cherry-pick" configuration is when there are unfinished items that should not be exported to the next system.

3. Have the admin go to Configuration Manager of App Setup, choose Sapio Configurations, and export relevant Sapio Configurations

Tip : Sapio Configurations export custom config including all customizations under Configuration Manager. These include Processes, Storage configuration settings, Pick list contents (including Sample Types), etc.

Note: In the event of an import failure, or unacceptable changes in configuration (such as uploading the wrong file), there is an option to restore from backup. ***There is only one backup at any point in time.***

Conversely, importing to a new environment should include...

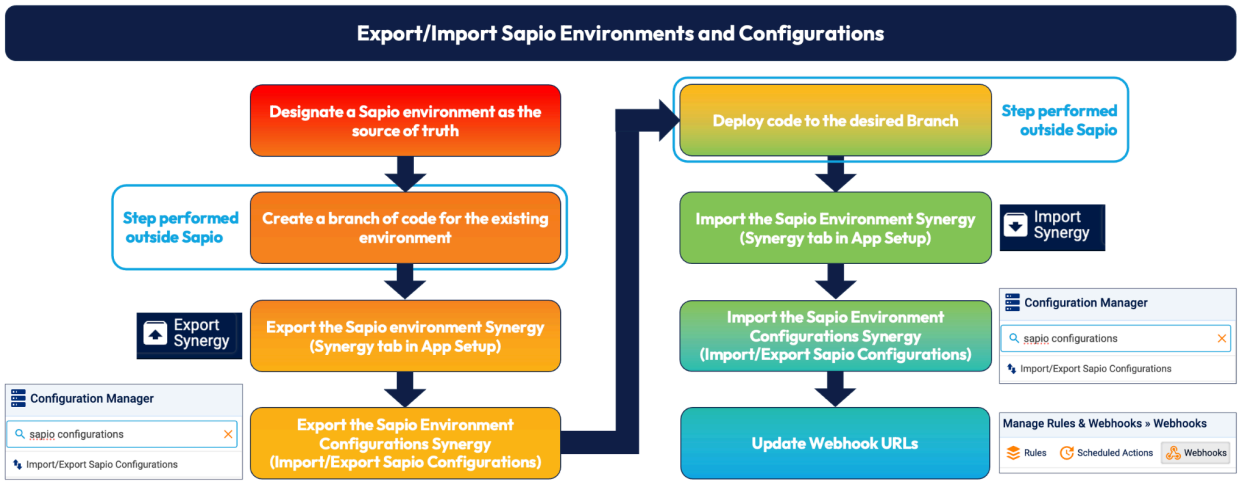
1. Deploying and code to the appropriate branch outside of Sapio

2. Have an admin go to App Setup > Import Synergy and selecting the .ssg file generated from the Export Synergy button

3. Have the admin go to Configuration Manager of App Setup, choose Sapio Configurations, and import previously exported Sapio Configurations

4. **Optional :** If using Webhooks, have the Admin navigate to 'Manage Rules and Webhooks' of App Setup and replace the URLs for the imported webhooks with the new environment's base URL

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**



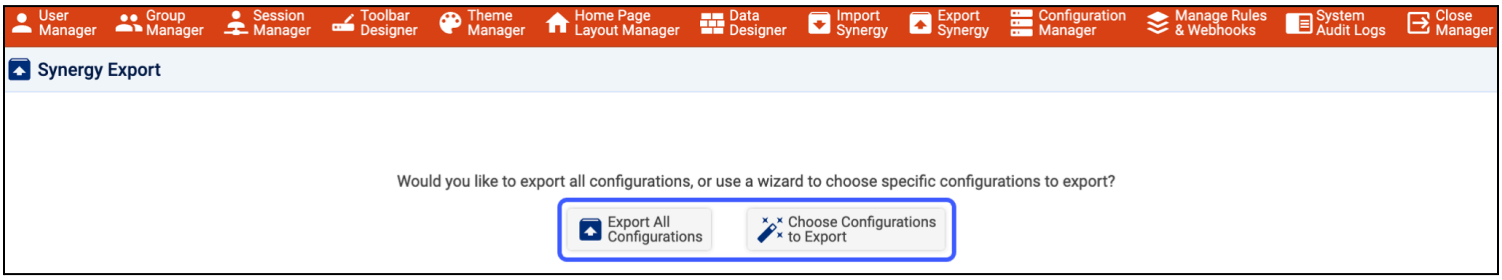
Export General System Settings with a Synergy

The Export Synergy tool in App Setup allows an admin to build and export a synergy file that can be used to transition the application's design and data structure settings to another application. Export Synergy is mainly used to send and receive design changes during development or to transition between development and production environments.

Note : Export Synergy does **NOT** transfer record data. Export Synergy also does **NOT** transfer user assignments to groups, but does copy existing groups.

1. From App Setup, click 'Export Synergy'

2. Select 'Export All Configurations' or 'Choose Configurations to Export'



3. **Optional :** If 'Choose Configurations to Export' was selected, this opens the first of several submenus where users can select individual configuration items to export. For each submenu, select available settings on the right to export

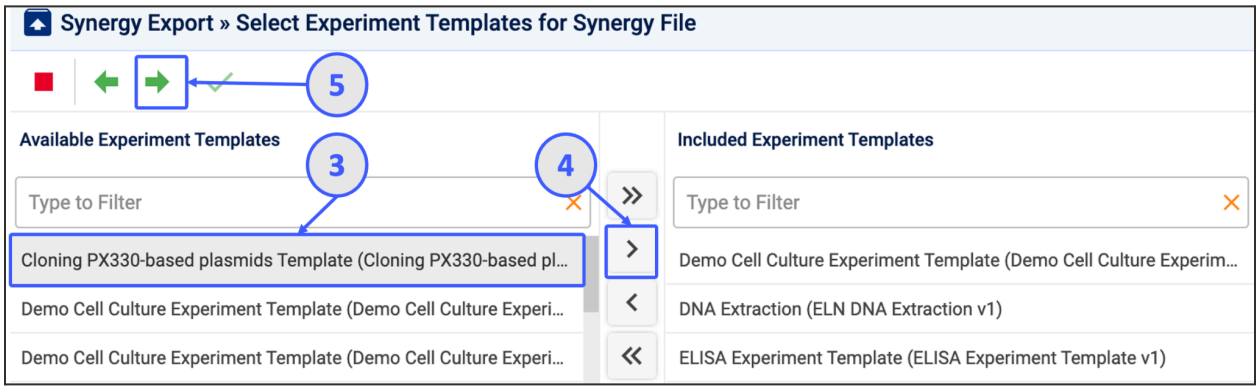
400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

4. Click the > arrow to move the setting from 'Available' to 'Included'

Tip : Some selected components may trigger the inclusion of other components for migration. For example, a Data Type with a field backed by a configured picklist, when selected, will automatically add the picklist to the Included Field Sets. Included downstream components can be manually removed in the same window shown above. Pay close attention to what data is being included in each component view, even if you don't intend to move over any settings from that component.

5. Select the green arrow to progress to the next group of settings

6. Repeat steps 3-5 until the green checkmark is enable



7. Click the green checkmark when finished

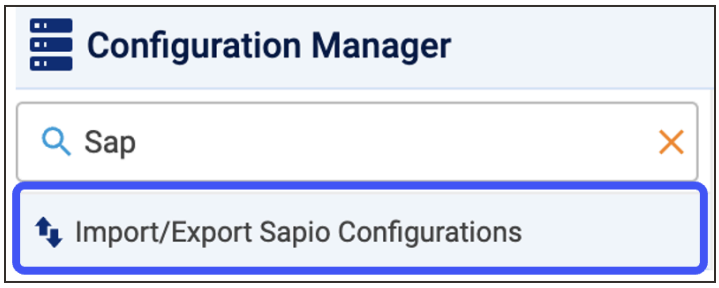
Export Sapio Configurations

Sapio Configurations should be exported as a part of migrating your custom configurations to other environments. This normally follows exporting a synergy if migrating all system changes, or after modifying individual configurations under configuration manager in app setup. To export Sapio Configurations...

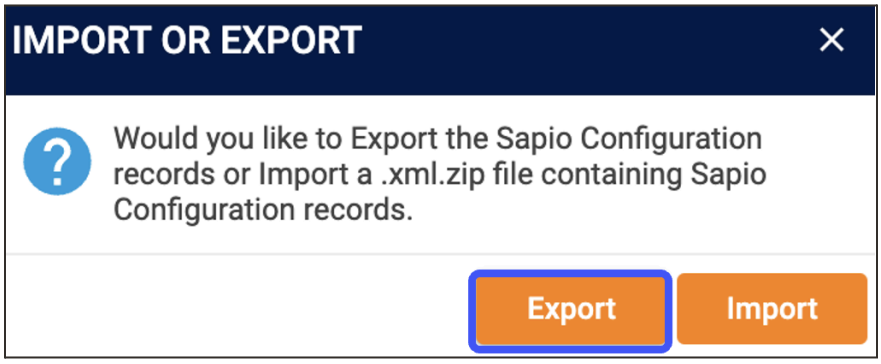
1. Navigate to App Setup and click the 'Configuration Manager' button

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

2. Filter and choose 'Import/Export Sapio Configurations'



3. Select 'Export'. A file called "SapioConfigurationsExport.xml.zip" will be downloaded.



Note : The downloaded zip file should remain a zip and NOT be extracted

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

Import General System Settings with a Synergy

Import Synergy allows an admin to upload a synergy file that can be used to move the application's design and data structure settings to another application. Import Synergy is mainly used to send and receive design changes during development or to transition between Test/QA environments and production environments.

If existing configurations are present, they can be overwritten. For example, if a Data Type has had a specific Data Field added to its Layout in the test environment, and the synergy file has that Data Type selected, importing the synergy can overwrite the old Data Type Layout.

WARNING : Allowing an Import Synergy to overwrite Sapio settings **CANNOT BE UNDONE**. It is advised to back up existing data or create a synergy of the current settings (export the existing configuration and name it appropriately) before deploying major changes.

To import in the new environment...

1. From App Setup, click 'Import Synergy'

2. **Optional :** Select any checkboxes you want to apply to your import data

   - **Overwrite existing Data Type, Data Field, and Pick List settings with settings from synergy**

     - **Checked :** The incoming synergy will overwrite existing settings as well as create new ones

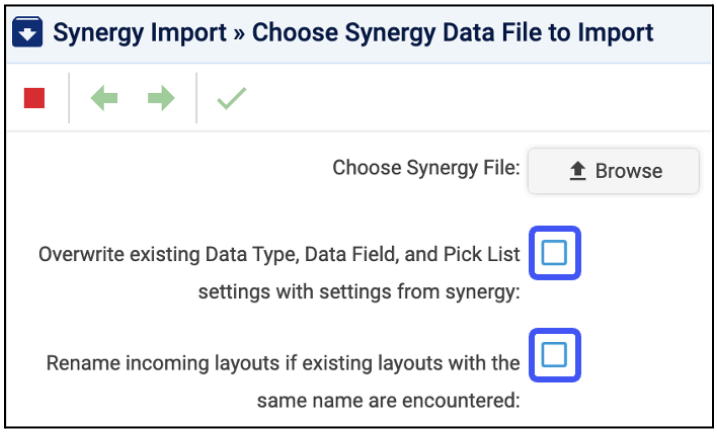
     - **Unchecked :** The incoming synergy will only add the import settings as new entries

   - **Rename incoming layouts if existing layouts with the same name are encountered**

     - **Checked :** The incoming synergy will rename incoming layouts if layouts with the same name already exist in the Sapio environment

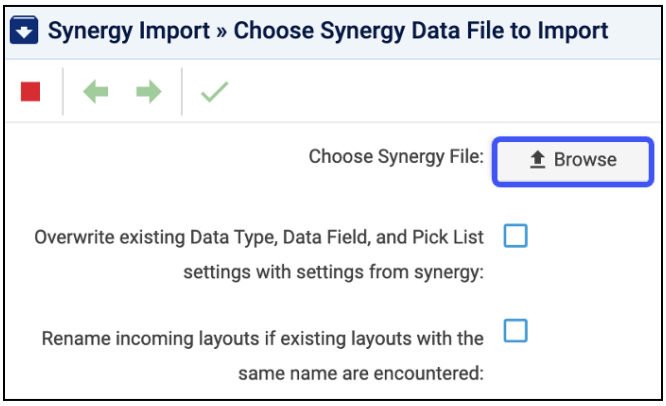
     - **Unchecked :** The incoming synergy will overwrite existing layouts with incoming layouts if they have the same name

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**



Tip : Report Builder templates in a synergy cannot be added to a system if a Template ID in the report builder already exists, even if the 'overwrite existing data' checkbox has been checked. If you wish to import a Report Builder Template when one of the same ID already exists, change the Template ID of the existing template and then load the synergy.

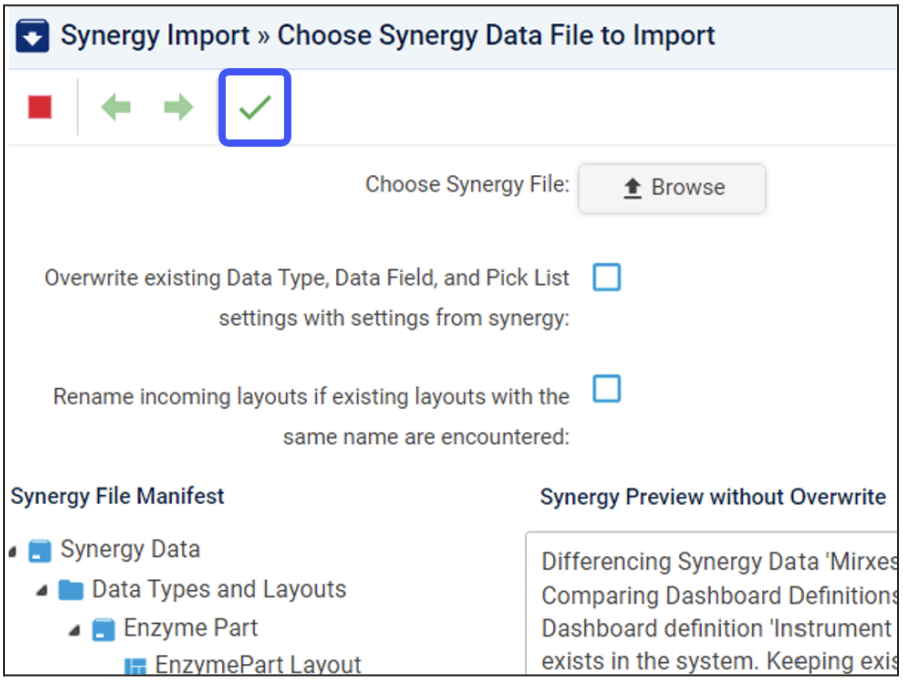
3. Click 'Browse'



4. Drag and Drop the file into the 'Choose a Synergy file to import' window, or click 'Browse Files' and select the .ssg file to import

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

5. After the file successfully loads, click the green checkmark at the top to save



Tip: Look for any error messages in the synergy logs. Sometimes due to versioning, errors may occur. Note any errors. For this reason, it is highly recommended to save the synergy log for reference or evidence of deployment.

Import Sapio Configurations

Sapio Configurations can be imported as a part of migrating your custom configurations to other environments. This normally follows importing a synergy if migrating all system changes, or after modifying individual configurations under configuration manager in app setup.

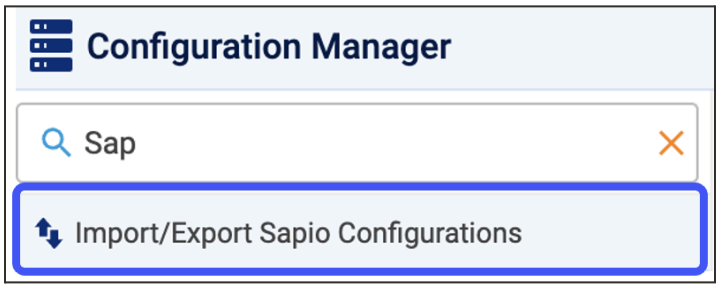
WARNING : Importing **will delete** the current/existing Sapio Configurations records. This **CANNOT BE UNDONE** or reverted, **even if you do not select a file to import, the records will still be removed.**

To import Sapio Configurations…

1. Navigate to App Setup and click the 'Configuration Manager' button

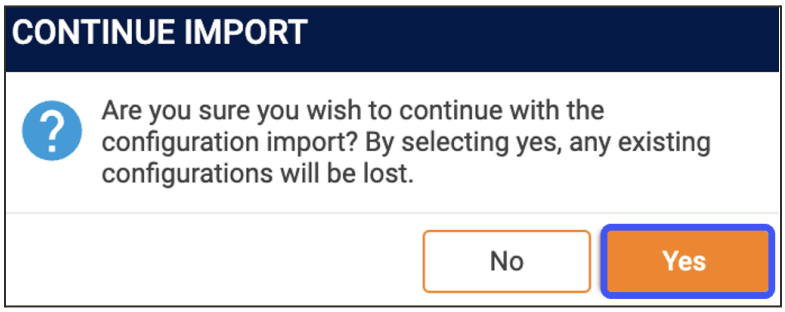
400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

2. Filter and choose 'Import/Export Sapio Configurations'

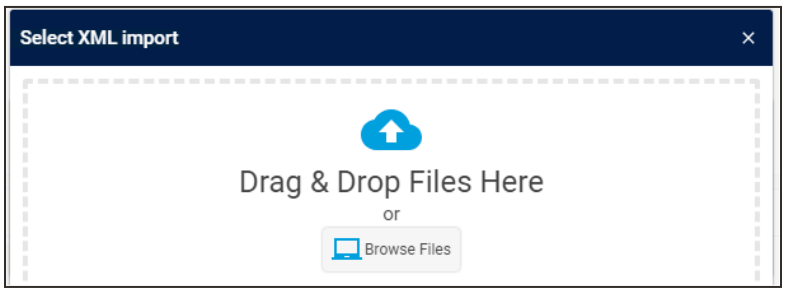


3. Select 'Import'

4. Confirm the import by selecting 'Yes'



5. Upload the previously downloaded **"SapioConfigurationsExport.xml.zip"** file



400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

Update Webhook Base URLs following synergy import of Webhooks

Upon successfully migrating their webhooks to a new environment with a synergy, users will need to update the base URLs for any webhook that used the previous environment's domain. For example, say you were moving from the dev environment to the test environment. These 2 environments have unique domains.

In this example...

Dev Environment: http//Sapio_dev_environment.exemplareln.com
Test Environment: http//Sapio_test_environment.exemplareln.com

Therefore, when migrating settings between these environments, any webhooks using the base url of "http//Sapio_dev_environment.exemplareln.com" would need to be changed to "http//Sapio_test_environment.exemplareln.com".

To update Webhooks base URLs...

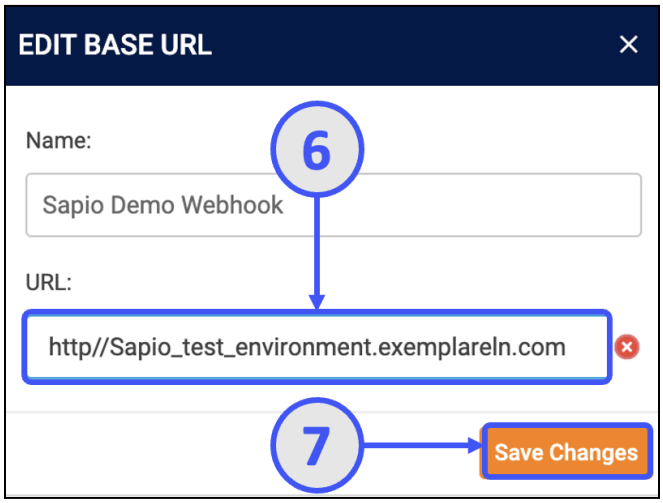
1. From App Setup, click 'Manage Rules and Webhooks'
2. Select the 'Webhooks' tab
3. Click the 'Base URL' subtab to see all webhooks
4. Filter the webhooks for the previous environments domain to remove non-impacted webhooks
5. Click edit for the webhook in the list to replace the base URL

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**

6. Click the 'URL' box, then copy and paste the new base url into the space

Tip: You can find the domain of your current Sapio environment on your web browser in the URL bar when you first login.

7. Click 'Save Changes'



8. Repeat steps 5-7 for all impacted webhooks

Note : These base urls only need to be changed for webhooks that reference the old Sapio environment domain.

Alternatively, if you make a list of all impacted webhooks and record the webhook names, you can replace the base URLs by going to App Setup > Manage Rules and Webhooks > Webhooks > WEBHOOKS > Replace Domain or Base URL and selecting the name of the impacted webhook.

400 E. Pratt Street Suite 800
Baltimore, MD 21202
(410) 800-4620
This document is **RESTRICTED**